

Sooyong Lee, Ph.D.

Psychometrician

WIDA

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PROFESSIONAL SUMMARY

As a psychometrician, I specialize in applying contemporary statistical theories and techniques to my work. My research interests center around developing and utilizing statistical models, including structural equation modeling (SEM), multilevel modeling (MLM), item response theory (IRT), and mixture modeling. In my recent work, I have placed particular emphasis on Bayesian estimation as a means of modeling the relationships between variables.

CONTACTS OF REFERENCES

1. Tiffany A. Whittaker

Professor
The University of Texas at Austin
Department of Educational Psychology
Quantitative Methods
Email: t.whittaker@austin.utexas.edu
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2. Seung W. Choi

Professor
The University of Texas at Austin
Department of Educational Psychology
Quantitative Methods
Email: schoi@austin.utexas.edu
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3. Cory L. Cobb

Assistant Professor
Texas A&M University
School of Public Health
Department of Health Behavior
Email: cory.cobb@tamu.edu
Phone: 501-749-3382

EDUCATION

- 2017-2023** Ph.D., Major in Quantitative Methods in Educational Psychology
The University of Texas at Austin, Austin, Texas
- Dissertation: Development of a Method for Addressing Various Censoring Effects in a Latent Growth Curve Modeling Framework. Advisor: Tiffany Whittaker
- 2014-2017** M.A., Major in Educational Statistics and Measurement
Korea University, Seoul, Republic of Korea
- Thesis: Exploration of LCA conditions in Latent Class with Low Proportion. Advisor: Sehee Hong
- 2007-2014** B.A. Major in International Studies
Hanyang University, Seoul, Republic of Korea
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PUBLICATIONS

In Preparation

- 35 Sooyong Lee** (2026). How to Generate Data for Simulation-based Power Analysis. In preparation.
- 34 Sooyong Lee**, Hacer Karamese, & Kyungwon Bishop (2026). Optimal Vertical Scaling Design for Short Tests with Polytomous Items. In preparation.

- 33 **Sooyong Lee**, Kahyun Lee, & Allison Masters (2026). Validity Study for AI-generated Sample and Responses for gender stereotype in STEM. In preparation.
- 32 **Sooyong Lee**, Hacer Karamese, & Kyungwon Bishop (2026). Longitudinal Trajectories of English-Language Proficiency for Students With Significant Cognitive Disabilities: A Latent Growth-Curve Approach. In preparation.

Under Review

- 31 **Sooyong Lee**, Myoungsung Lee, & Sehee Hong (2026). Using Vertical Scaling for Second-order Latent Growth Curve Models.. *Structural Equation Modeling: A Multidisciplinary Journal*. Under review.
- 30 **Sooyong Lee**, Cory L. Cobb, & Soyoung Lee (2026). Comparing Methods for Testing Mediation Effects Involving Nominal Mediators: A Simulation and Empirical Study. *Multivariate Behavioral Research*. Under review.
- 29 Cory L. Cobb, **Sooyong Lee** (2026). Cultural Stress, Psychopathology, and Wellbeing among Latinos: A Two Continua Model Approach. Under review.
- 28 Cory L. Cobb, **Sooyong Lee**, Lawrence Watkins, Seth J. Schwartz, Duyen J. Vo, & Seo Woo Lee (2026). Dispositional Gratitude as a Psychological Resource: Differential Links Between Self-Critical Rumination, Well-Being, and Psychopathology in Hispanics. *Journal of Personality*. Under review.
- 27 Kahyun Lee, **Sooyong Lee**, & Allison Master (2026). Understanding Adolescents'Pattern of Change in Growth Mindsets about Math. *Journal of Educational Psychology*. Under review.
- 26 Cory L. Cobb, **Sooyong Lee**, & Tae Kyoung Lee (2026). Psychopathology Typologies and their Correlates among Latinos: A Latent Profile Analysis. *International Journal of Social Psychiatry*. Under review.
- 25 Kahyun Lee, **Sooyong Lee**, Seungmin Jahng & Allison Master (2026). An Intensive Longitudinal Study of Emotion, Growth Mindset, and Class Experiences. *Emotion*. Under review.
- 24 Cory L. Cobb, **Sooyong Lee**, Lawson Kami, Schwartz Seth, Eddy J. Mark, & Martinez Charles (2026). The Cultural Continuity Scale: Cross-Validated Item Response Theory Development of a Measure of Perceived Cultural Stability in a Racially and Ethnically Diverse U.S. Sample. *Assessment*. Under review.

2026

- 23 Rene M. Dailey, Lingzi Zhong, Sarah Varga, & **Sooyong Lee** (2026). Testing A Comprehensive Model of Post-Dissolution Distress: Predicting Initial and Over Time Changes in Distress Following Romantic Dissolution among U.S. Adults. *Personal Relationships*. In print (Accepted).
- 22 **Sooyong Lee** & Soyoung Kim (2026). Comparing Bayesian Random Coefficient Prediction and Latent Interaction Models for Multilevel Moderated Mediation. *Frontier in Psychology*. DOI:DOI: 10.3389/fpsyg.2025.1543330. In print.

2025

- 21 Jiwon Kim, **Sooyong Lee**, Suyoung Kim, & Brian Keller (2025). Tutorial for Bayesian Multilevel Structural Equation Modeling using Blimp. *Structural Equation Modeling: A Multidisciplinary Journal*. 1-13. DOI:https://doi.org/10.1080/10705511.2025.2526035.
- 20 **Sooyong Lee**, Adam Sales, Hyeon-Ah Kang, & Tiffany A. Whittaker (2025). Model Estimation Approaches for Fully-latent Principal Stratification with Small Samples. *Structural Equation Modeling: A Multidisciplinary Journal*. 32(2), 251-263. DOI:https://doi.org/10.1080/10705511.2024.2402331.

- 19 **Sooyong Lee**, Kahyun Lee, & Hanjo Kim (2025). Mediation Analysis with an Event-history Mediator: A Simulation Study and Empirical Data Illustration. *Structural Equation Modeling: A Multidisciplinary Journal*. 1-11. DOI:<https://doi.org/10.1080/10705511.2025.2478587>.
- 18 **Sooyong Lee** & Tiffany A. Whittaker (2025). Development of a Method for Handling Doubly-Censored Data in a Latent Growth Curve Modeling Framework. *Multivariate Behavioral Research*. 1-17. DOI:<https://doi.org/10.1080/00273171.2025.2478071>.
- 17 **Sooyong Lee**, Adam Sales, Hyeon-Ah Kang, & Tiffany A. Whittaker (2025). Fully Latent Principal Stratification With Measurement Models. *Journal of Educational and Behavioral Statistics*. DOI:<https://doi.org/10.3102/10769986251321428>.
- 16 Garron Gianopulos, Jonghwan Lee, Sangdon Lim, Luping Niu, **Sooyong Lee**, & Seung W. Choi (2025). The Impact of Item Bank Size and Item Bank Distribution on Student Ability Estimates for a Hybrid Interim-Summative CAT. *Journal of Computerized Adaptive Testing*. 12, 54-87. DOI:DOI 10.7333/2502-1201054.
- 15 Lee, J, Lim, S., Schneider, C, Gianopulos, G, Niu, L., **Sooyong Lee**., & Choi, S. W. (2025). The Impact of Item Bank Transition Rules on Student Ability Estimates and Achievement Level Classifications. *Journal of Computerized Adaptive Testing*. 12, 88-122. DOI:DOI 10.7333/2502-1201088.
- 14 **Sooyong Lee**, Suyoung Kim, & Seung W. Choi (2025). Anchor Detection Strategy in Moderated Non-linear Factor Analysis for Differential Item Functioning (DIF). *Applied Psychological Measurement*. DOI:10.1177/01466216251401206. In print.

2024

- 13 Suyoung Kim, **Sooyong Lee**, Jiwon Kim, & Tiffany A. Whittaker (2024). Investigating Latent Interaction Effects in Multiple-Group Analysis in the Structural Equation Modeling Framework. *Structural Equation Modeling: A Multidisciplinary Journal*. 31(6), 1043-1055. DOI:<https://doi.org/10.1080/10705511.2024.2363827>.
- 12 **Sooyong Lee**, Suhwa Han, & Seung W. Choi (2024). A Bayesian Moderated Nonlinear Factor Analysis Approach for DIF Detection under Violation of the Equal Variance Assumption. *Journal of Educational Measurement*. 62(2), 303-324. DOI:<https://doi.org/10.1111/jedm.12388>.
- 11 **Sooyong Lee**, Kahyun Lee, & Kejin Lee (2024). Discrete-time Survival Analysis Incorporating Time Structure in Developmental Research. *Structural Equation Modeling: A Multidisciplinary Journal*. DOI:<https://www.tandfonline.com/doi/abs/10.1080/10705511.2024.2432598>. In print.

2023

- 10 **Sooyong Lee**, Adam Sales, Hyeon-Ah Kang, & Tiffany A. Whittaker (2023). Fully Latent Principal Stratification: Combining PS with Model-Based Measurement Models.. *The Annual Meeting of the Psychometric Society*. pp. 287-298. DOI:<https://doi.org/10.1007/978-3-031-27781-8> (Conference proceedings).
- 9 **Sooyong Lee** & Tiffany A. Whittaker (2023). Parallel Process Latent Growth Modeling with Multivariate Confounders/Suppressors. *Structural Equation Modeling: A Multidisciplinary Journal*. 30(2), 284-295. DOI:<https://doi.org/10.1080/10705511.2022.2133774>.
- 8 **Sooyong Lee**, Kahyun Lee, & Tiffany A. Whittaker (2023). Application of Associative Discrete-Time Survival Analysis Using Latent Transition Specification. *Structural Equation Modeling: A Multidisciplinary Journal*. 30(5), 831-842. DOI:<https://doi.org/10.1080/10705511.2022.2138892>.
- 7 Sales, A., **Sooyong Lee**, Whittaker, T., & Kang, H. A (2023). Fully-Latent Principal Stratification for Modeling Complex Program Implementation Data. *Society for Research on Educational Effectiveness*. DOI:<https://eric.ed.gov/?id=ED659422> (Report).

2022

- 6 **Sooyong Lee** & Tiffany A. Whittaker (2022). Comparison of Three Approaches to Class Enumeration in Growth Mixture Modeling when Time Structures are Variant Across Latent Classes. *Structural Equation Modeling: A Multidisciplinary Journal*. 29, 23-35. DOI:<https://doi.org/10.1080/10705511.2021.1956320>.
- 5 **Sooyong Lee**, Suhwa Han, & Seung W. Choi (2022). DIF Detection With Zero-Inflation Under the Factor Mixture Modeling Framework. *Educational and Psychological Measurement*. 82-4, 678-704. DOI:<https://doi.org/10.1177/00131644211028995>.
- 4 Seung Choi, Sangdon Lim, Luping Niu, **Sooyong Lee**, Christina M. Schneider, Jay Lee, & Garron J. (2022). maat: An R Package for Multiple Administrations Adaptive Testing. *Applied Psychological Measurement*. 46-1, 73-74. DOI:<https://doi.org/10.1177/01466216211049212>.

2021

- 3 Cory L. Cobb, Charles R. Martínez, Jr., **Sooyong Lee**, Tae Kyoung Lee, & Elma I. Lorenzo-Blanco (2021). Acculturation trajectories differ by youth age at arrival and time in residency among Latino immigrant families in a US emerging immigrant context. *International Journal of Intercultural Relations*. 81, 79-93. DOI:[10.1016/j.ijintrel.2021.01.005](https://doi.org/10.1016/j.ijintrel.2021.01.005).

2020 and earlier

- 2 Eun Soo Lee, **Sooyong Lee**, & Sehee Hong (2016). The Effects of Community Child-Center Service on School Adjustment Change Trajectory. *Korean Journal of Social Welfare Research*. 50, 59-85.
- 1 **Sooyong Lee**, Eun Soo Lee, & Sehee Hong (2016). Effects of adolescent's career development on labor market outcomes: Applying the two-part growth mixture model. *Journal of Vocational Education & Training*. 19(3), 137-162.

NATIONAL AND INTERNATIONAL PRESENTATIONS

Presentations

- 10 Kei Bishop, Hacer Karamese, Syed Abdul Hadi, & **Sooyong Lee** (2025-04). *Two rating scale Rasch model linking in a large-scale language assessment: psychometric and content reviews*. International Objective Measurement Workshop, Denver, USA.
- 9 Syed Abdul Hadi, Kei Bishop, Hacer Karamese, & **Sooyong Lee** (2025-04). *Automated Plagiarism Detection in Spoken Responses using Linguistic and Acoustic Features*. National Council on Measurement in Education (NCME), Denver, USA.
- 8 Syed Abdul Hadi, Kei Bishop, Hacer Karamese, & **Sooyong Lee** (2024-09). *Automated Essay Scoring: Syntactic and Semantic Feature Engineering for Interpretable and Content Aware System*. International Association for Computerized Adaptive Testing (IACAT), Seoul, South Korea.
- 7 Kei Bishop, Hacer Karamese, **Sooyong Lee**, & Syed Abdul Hadi (2024-09). *Comparing Vertical Scaling Methods in Multistage Adaptive Testing using Simulation and Operational Data on WIDA's Language Assessments*. International Association for Computerized Adaptive Testing (IACAT), Seoul, South Korea.
- 6 **Sooyong Lee**, Adam Sales, Hyeon-Ah Kang, & Tiffany A. Whittaker (2024-04). *Multidimensional Fully Latent Principal Stratification*. American Educational Research Association (AERA), Philadelphia, USA.

- 5 Tiffany Whittaker, Hyeon-Ah Kang, **Sooyong Lee**, Adam Sales (2023-04). *Bayesian and Maximum Likelihood Estimation in Fully Latent Principal Stratification*. National Council on Measurement in Education (NCME), Chicago, USA.
- 4 **Sooyong Lee**, Adam Sales, Hyeon-Ah Kang, & Tiffany A. Whittaker (2022-07). *Fully-Latent Principal Stratification*. International Meeting of the Psychometric Society (IMPS), Bologna, Italy.
- 3 Garron Gianopoulos, Jay Lee, Sangdon Lim, Luping Niu, **Sooyong Lee**, & Seung W. Choi (2022-04). *The Impact of Item Pool Size and Item Pool Distribution on Student Ability Estimates for a Hybrid Interim-Summative CAT*. National Council on Measurement in Education (NCME), San Diego, USA.
- 2 Jay Lee, M. Christina Schneider, Garron Gia, Luping Niu, **Sooyong Lee**, Sangdon Lim, & Seung W. Choi (2022-04). *The Impact of item bank transition rules on student ability estimates and achievement level classifications*. National Council on Measurement in Education (NCME), San Diego, USA.
- 1 Daniel Lewis & **Sooyong Lee**, Seung Choi (2021-04). *An Investigation of Two Embedded Standard Setting Cut Score Estimation Algorithms: ESS-Count & ESS-Weight..* National Council on Measurement in Education (NCME), Baltimore, MD, USA.

Posters

- 9 **Sooyong Lee**, Kei Bishop, & Hacer Karamese (2026-04). *Optimal Vertical Scaling Design for Short Tests with Polytomous Items*. National Council on Measurement in Education (NCME), LA, USA.
- 8 **Sooyong Lee**, Kei Bishop, Hacer Karamese, & Syed Abdul Hadi (2025-04). *Censored Confirmatory Factor Analysis for Non-Normal Likert-Type Data in Language*. National Council on Measurement in Education (NCME), Denver, USA.
- 7 Hacer Karamese, Kei Bishop, **Sooyong Lee**, & Syed Abdul Hadi (2025-04). *An Empirical Comparison of Linking Methods in Multistage Adaptive Test*. National Council on Measurement in Education (NCME), Denver, USA.
- 6 **Sooyong Lee**, Kahyun Lee, & Allison Master (2024-04). *Exploring the dynamics of daily experiences, emotions, and depression among Asian American students: The moderating role of growth mindsets..* American Educational Research Association (AERA), Philadelphia, USA.
- 5 **Sooyong Lee**, Suhwa Han, & Seung W. Choi (2021-04). *DIF Detection with Zero-Inflation under the Mixture Modeling Framework*. American Educational Research Association (AERA), Virtual.
- 4 **Sooyong Lee** & Suhwa Han (2020-07). *Item selection and trait estimation methods in CAT with item leakage*. International Meeting of the Psychometric Society (IMPS), Maryland, USA.
- 3 **Sooyong Lee** & Tiffany A. Whittaker (2020-04). *Comparison of Three Approaches to Class Enumeration in Growth Mixture Modeling when Time Structures are Variant Across Latent Classes*. American Educational Research Association (AERA), Virtual (Cancelled).
- 2 **Sooyong Lee** & Sehee Hong (2018-07). *Exploration of LCA conditions in Latent Class with Low Proportion*. International Meeting of the Psychometric Society (IMPS), New York, USA.
- 1 Kejin Lee, **Sooyong Lee**, & Youngri Lee (2018-05). *An Alternative Approach to Discrete-Time Survival Analysis for Meaningful Interpretation*. Modern Modeling Methods conference (MMM), Connecticut, USA.

RESEARCH

Psychometrician	WIDA - Wisconsin Center for Education Research - Psychometrics for K-12 students Language tests	2024/4-present
Post-doc Fellow	FLPS project - Institute of Educational Sciences (IES) grant \$891,895 2021–2025 PI: Tiffany Whittaker - Development of 'flps' package for Fully-latent Principal Stratification	2023/9-2024/3
Research Assistant	FLPS project - Institute of Educational Sciences (IES) grant \$891,895 2021–2025 PI: Tiffany Whittaker - Development of 'flps' package for Fully-latent Principal Stratification	2021/9-2023/8
Intern	Embedded Standard Setting (ESS) - Creative Measurement Solutions LLC - Internship for developing ESS package for Embedded Standard Setting	2021/6-2021/8
Intern	Embedded Standard Setting (ESS) - Creative Measurement Solutions LLC - Internship for developing Shiny app for Embedded Standard Setting	2020/6-2020/8
Research Assistant	'maat' package development project - Grant from Northwest Evaluation Association (NWEA) - Development of MAAT package for Computerized Adaptive Testing for Northwest Evaluation Association (NWEA)	2019/9-2020/5
Research Assistant	Survey of COE Full-Time Tenured and Tenure-Track Faculty - College of Education - Data analysis for surveys of graduate programs of education at the University of Texas at Austin	2019/9-2019/12
Research Assistant	President's Award for Global Learning (PAGL) - President's Award for Global Learning (PAGL) - Designing a culturally appropriate mobile application to increase mental health literacy and self-care habits among South Korean young adults	2019/1-2019/8
Research Assistant	CGS PhD Career Pathways project - College of Education - Data analysis for surveys to collect data on the career pathways of past and present STEM and humanities PhD students.	2018/9-2019/5

TEACHING

Teaching Assistant

University of Texas at Austin

EDP 380C: Structural Equation Modeling
Professor Tiffany Whittaker
University of Texas at Austin

2021/1-2021/5

EDP 380C: Structural Equation Modeling Professor Tiffany Whittaker Univeristy of Texas at Austin	2020/1-2020/5
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Informal Special Talks

Introduction to Basic Linear Algbra for Statistics Link to Materials Conducted foundational training sessions on linear algebra for advanced statistics	2025/05
Introduction to Latent Class Analysis Link to Materials Conducted introductory training sessions on latent class analysis	2024/09
Shiny app development using flexdashboard Link to Materials Conducted introductory training sessions on building Shiny applications in R	2023/01
visulization with ggplot2 Link to Materials Conducted hands-on training sessions on data visualization using the ggplot2 package	2022/06
LGM using lavaan package Link to Materials Conducted introductory training sessions on latent growth model analysis using the lavaan package	2021/06
Applied Research using R Link to Materials Conducted introductory training sessions on conducting applied research using R	2021/05
Simulation Study using R package development Link to Materials Conducted introductory training sessions on statistical simulation analysis using R packages	2021/05
Data Generation in R Link to Materials Conducted training sessions on data generation techniques for statistical simulation	2021/03

PROGRAMS

R package	GRshiny package https://sooyongl.github.io/GRShiny/	2023
R package	clavaan package https://github.com/sooyongl/clavaan	2023
Shiny application	GRshiny: Graded response model https://sooyonglee.shinyapps.io/GRShiny/	2022
R package	Embedded Standard Setting https://sooyongl.github.io/EmStanS/	2022
R package	maat package https://choi-phd.github.io/maat/	2022

R package	flps package https://cran.r-project.org/web/packages/flps/index.html	2022
Shiny application	Embeded Standard Setting	2021

SERVICE

Invited Reviewer	BMC Medical Research Methodology	2026
Invited Reviewer	Structural Equation Modeling: A Multidisciplinary Journal	2024
Invited Reviewer	Qeios	2024
Reviewer	International Journal of Intercultural Relations	2023/7 - present